

分式（一）

1. 当 x 取哪些值时，下列分式为 0？

(1) $\frac{2x-1}{x+3}$

(2) $\frac{8x}{x^2+8}$

(3) $\frac{x^2-2x-3}{(x+1)(x+2)}$

(4) $\frac{|x|-6}{x^2-5x-6}$

2. 当 x 取哪些值时，下列分式无意义？

(1) $\frac{1}{1+\frac{1}{1+x}}$

(2) $\frac{x^2-x-2}{|x|+2}$

3. 化简：

(1) $\left(1-\frac{1}{1-x}\right)\left(\frac{1}{x^2}-1\right)$

(2) $\left(\frac{x+2}{x^2-2x}-\frac{x-1}{x^2-4x+4}\right)\div\frac{x-4}{x}$

(3) $\frac{3-m}{2m-4}\div\left(m+2-\frac{5}{m-2}\right)$

(4) $1-\frac{a-b}{a+2b}\div\frac{a^2-b^2}{a^2+4ab+4b^2}$

4. 化简下列分式:

$$(1) \frac{1}{a-b} + \frac{1}{a+b} + \frac{2a}{a^2+b^2} + \frac{4a^3}{a^4+b^4}$$

$$(2) \frac{x^3-1}{x^3+2x^2+2x+1} + \frac{x^3+1}{x^3-2x^2+2x-1} - \frac{2(x^2+1)}{x^2-1}$$

5. (1) 若 $\frac{a}{b}=2$, 求 $\frac{a^2-ab+b^2}{a^2+b^2}$ 的值.

(2) 若 $\frac{x}{2} = \frac{y}{3} = \frac{z}{4}$, 求 $\frac{xy+yz+zx}{x^2+y^2+z^2}$ 的值.

6. 已知 $\frac{x}{y+z+u} = \frac{y}{z+u+x} = \frac{z}{u+x+y} = \frac{u}{x+y+z}$, 求 $\frac{x+y}{z+u} + \frac{y+z}{u+x} +$

$\frac{z+u}{x+y} + \frac{u+x}{y+z}$ 的值.

7. 若 $x+y+z \neq 0, x+y \neq 0, y+z \neq 0, z+x \neq 0$, 且 $a = \frac{x}{y+z}, b = \frac{y}{x+z},$

$c = \frac{z}{x+y}$, 证明: $\frac{a}{a+1} + \frac{b}{b+1} + \frac{c}{c+1} = 1.$

8. 已知: $\frac{x}{y} = \frac{a}{b}$, 证明: $\frac{x^2 + a^2}{x + a} + \frac{y^2 + b^2}{y + b} = \frac{(x + y)^2 + (a + b)^2}{x + y + a + b}$.

9. 证明:

$$\frac{b-c}{(a-b)(a-c)} + \frac{c-a}{(b-c)(b-a)} + \frac{a-b}{(c-a)(c-b)} = \frac{2}{a-b} + \frac{2}{b-c} + \frac{2}{c-a}.$$

10. 若 $x = \frac{a-b}{a+b}, y = \frac{b-c}{b+c}, z = \frac{c-a}{c+a}$, 证明: $(1+x)(1+y)(1+z) = (1-x)(1-y)(1-z)$.

11. 已知 $a^2 - 3a + 1 = 0$, 求 $\frac{2a^5 - 5a^4 + 2a^3 - 8a^2}{a^2 + 1}$ 的值 (选做).